When high flux helps single-pixel imaging: a contrast–dead-time operating map

[author block — user decision]

Abstract

Single-pixel imaging with photon-counting detectors is conventionally operated at low photon flux, keeping the detector dead-time loss below a few percent at the cost of long acquisition times. Here we report a contrast–dead-time operating map for high-flux single-pixel imaging: whether deliberately raising the operating flux into the strongly nonlinear dead-time regime helps or does not help is governed by the illumination-pattern bucket contrast, not by the dead-time correction alone. The nonlinearity is absorbed by a convex renewal quasi-likelihood (RQL) reconstruction with a deployment-legal, truth-free regularization rule, and a finite-window count-information analysis locates the principled boundary: the Fisher information of the integrated photon count possesses a principal ridge at the cube-root load $\rho^*(\nu) = (6\nu)^{1/3} - 2/3 + O(\nu^{-1/3})$ ($\nu = T/\tau$ dead-time slots per exposure). In a preregistered dense-pattern study (Study 1, 64^2 scenes, occupancy 50%), the time-to-quality endpoint was uninformative in 24/24 scenes (safe-range quality span below the preregistered 0.5 dB floor), while a small, uniformly positive fixed-dwell gain remained (+0.28 dB median, 24/24 scenes positive, bootstrap lower bound +0.23 dB) — the multiplex-limited regime. In a second preregistered study (Study 2, 32^2), spatial occupancy was varied at *equal mean detector load and equal incident dose across occupancies*, raising the measured bucket contrast from $C_u = 0.040$ (occupancy 50%) to 1.27 (raster). In Study 2, the preregistered acquisition-speed endpoint was formally negative because its breadth criterion failed (16/24 scenes with $S_{\text{gate}} > 1$, versus the required 18/24), although both effect-size criteria passed (median 6.80×; family-stratified bootstrap lower bound 2.84×). The preregistered fixed-dwell secondary passed (median +1.16 dB at the terminal dwell, 19/24 scenes positive, lower bound +0.72 dB). A post hoc descriptive family-level analysis found that the six success fractions were monotonically ordered by measured bucket contrast, consistent with—but not establishing—a contrast-controlled transition. Across preregistered no-gate occupancy controls, the fixed-dwell gain was nonmonotone and largest among the tested rungs at 3.1% occupancy, exposing a contrast–conditioning trade-off. The gain survived dynamic scattering with coefficient of variation 0.20. The resulting operating map opens up an avenue for photon-budget-limited and dwell-time-limited single-pixel systems.

1 Introduction

Single-pixel imaging (SPI) and computational ghost imaging reconstruct a scene from the temporal sequence of a single bucket detector's responses to structured illumination [1, 2, 3], and have proven particularly attractive where array sensors are unavailable or photon budgets are severe: low-light and photon-starved imaging, NIR/SWIR sensing, and imaging through scattering media such as turbid water [4, 5, 6]. In the photon-starved regime the bucket detector is a photon-counting device — a single-photon avalanche diode or photomultiplier — and image quality is set by how many photons can be counted within the available acquisition time.

Photon-counting detectors, however, are blind for a dead time τ after each registered count. The established response is to operate at low flux — keeping the dead-time duty $\rho = \lambda\tau$ at the few-percent level so that losses stay small and a linear correction suffices — and to pay for it with acquisition time, since the count rate is then a small fraction of the detector's $1/\tau$ ceiling. A substantial literature instead corrects or models the distortion at higher rates: dead-time compensation and high-flux operation in single-photon lidar [7, 8, 9, 10], count-rate corrections in photon-counting imaging [11], and dead-time-aware estimation bounds [12]. These works are correction-centric or timestamp-based; what remains unmapped is the operating question for the count-only bucket channel itself: at what flux, for which scenes, does raising the rate into the nonlinear regime actually help?

Previous studies established count-rate-dependent CRLB degradation, task-specific optimal operating fluxes, and high-flux recovery using time-resolved dead-time models [13, 14, 9, 7, 12]. Here, we study a different information bottleneck: a finite exposure from which only the integrated nonparalyzable count is retained. For this count-only observation, we derive a principal finite-window Fisher-information ridge and show that its optimal load obeys a cube-root law,

$\rho^*(\nu) = (6\nu)^{1/3} - 2/3 + O(\nu^{-1/3})$, arising from terminal detector-phase information loss.

Sparse illumination itself is not the claimed contribution of this work. Prior work has used sparse patterns to reduce detector count rates and alleviate pile-up [15, 16]. Here, instead, spatial occupancy is varied under equal mean detector load and equal incident dose, revealing a contrast-controlled boundary between multiplex-limited and photon-limited high-flux single-pixel imaging.

This work contributes: (i) an exact finite-window count-information analysis of the nonparalyzable channel, including a missing-information identity and the cube-root ridge of Eq. (5); (ii) a convex renewal quasi-likelihood reconstruction with a truth-free analytic regularization rule, validated against the exact likelihood; (iii) two preregistered simulation studies — dense-pattern and equal-load occupancy-ladder — whose endpoints, gates, scene lists, and display subjects were frozen in immutable commits before any confirmatory reconstruction existed, and whose outcomes are reported as frozen, including the formally negative primaries; and (iv) the resulting operating map with its computable engagement criterion. Figure 1 sets out the single-pixel imaging chain, the dead-time saturation mechanism, and the resulting contrast–dead-time operating map. All code, frozen preregistration documents, and per-cell evidence bundles are openly released.

2 Physical model and problem statement

A single-pixel detector with dead time τ observes, for illumination pattern a_i and scene $x \geq 0$, a Poisson arrival process of rate $\lambda_i = \Phi\, a_i^\top x + d$ during an exposure of duration $T = \nu\tau$; after each registered count the detector is blind for τ (non-paralyzable, active start). The recorded observation is the integrated count N_i alone, with exact law

$$P(N_i \geq m) = F_{\Gamma(m,\lambda_i)}\big(T - (m-1)\tau\big) = P\big(\mathrm{Pois}(z_m) \geq m\big), \quad z_m = \lambda_i\,(T - (m-1)\tau). \tag{1}$$

The per-slot load $\rho_i = \lambda_i \tau$ is the dead-time duty; the conventional regime keeps $\rho \lesssim 0.05$. Illumination ensembles used in this work: dense Bernoulli(1/2) patterns (Study 1 primary), complementary Hadamard-type pairs charged at $2M$ physical exposures and gamma-modulated patterns (supplementary ensemble sensitivity), and the Study-2 fixed-occupancy binary ladder of Section 6. Because patterns spread instantaneous load, deployment statements use the per-pattern 95th percentile of ρ rather than its mean; percentile ledgers for every cell are part of the released evidence bundles.

3 Renewal quasi-likelihood reconstruction

We reconstruct by

$$\hat{x} = \arg\min_{x\geq 0} \frac{1}{M}\sum_{i=1}^{M}\Big[(T - N_i\tau)\,\lambda_i(x) - N_i \log \lambda_i(x)\Big] + \lambda_{\mathrm{TV}}\,\mathrm{TV}(x), \tag{2}$$

with $\lambda_i(x) = \Phi a_i^\top x + d$. The data term is convex in λ (and in x through the affine map); its single-measurement stationary point $\hat{\lambda} = N/(T - N\tau)$ coincides with the classical dead-time rate correction, so the estimator's novelty resides not in the per-measurement formula but in the joint regularized inverse problem and the operating-point map it enables. At ceiling counts ($N\tau \geq T$) the objective is unbounded below in λ — the true identifiability boundary of count-only observation; the fraction of ceiling counts is recorded per cell. A full heteroscedastic Gaussian likelihood fails at high load through a variance-collapse non-coercivity (Supplement §S1).

3.1 Truth-free regularization selection and descriptive audit

The TV weight is set by a frozen analytic score-concentration rule with no per-scene tuning: $\lambda_{\mathrm{TV}} = c\,\sigma_g\sqrt{2\ln n}$ with global constant $c = 0.50$ frozen before any confirmatory data existed, and $\sigma_g = \Phi\sqrt{\kappa_A v_s/M}$ computed from the exact renewal per-count score variance v_s (no Gaussian approximation; the central limit theorem is quantitatively untrustworthy at short windows $\nu \leq 10$). All arms are scaled by the identical rule through their own κ_A.

The measurement audit is descriptive and does not constitute a model-adequacy test. All audit statistics are excluded from regularization selection, reconstruction, cell inclusion, and confirmatory pass/fail decisions. The refit-bootstrap reference distribution is conditional on a smoothed plug-in scene; its empirical tail ranks are therefore not calibrated model-class p-values for arbitrary scenes and can be anti-conservative in the upper tail when the true scene is rougher than the plug-in. Moreover, count-only observations collected at a single operating point cannot generally distinguish detector mean-response mismatch from radiometric rescaling of the scene. Detector-mismatch consequences are therefore quantified directly through the preregistered radiometric metrics in Section 6.7.

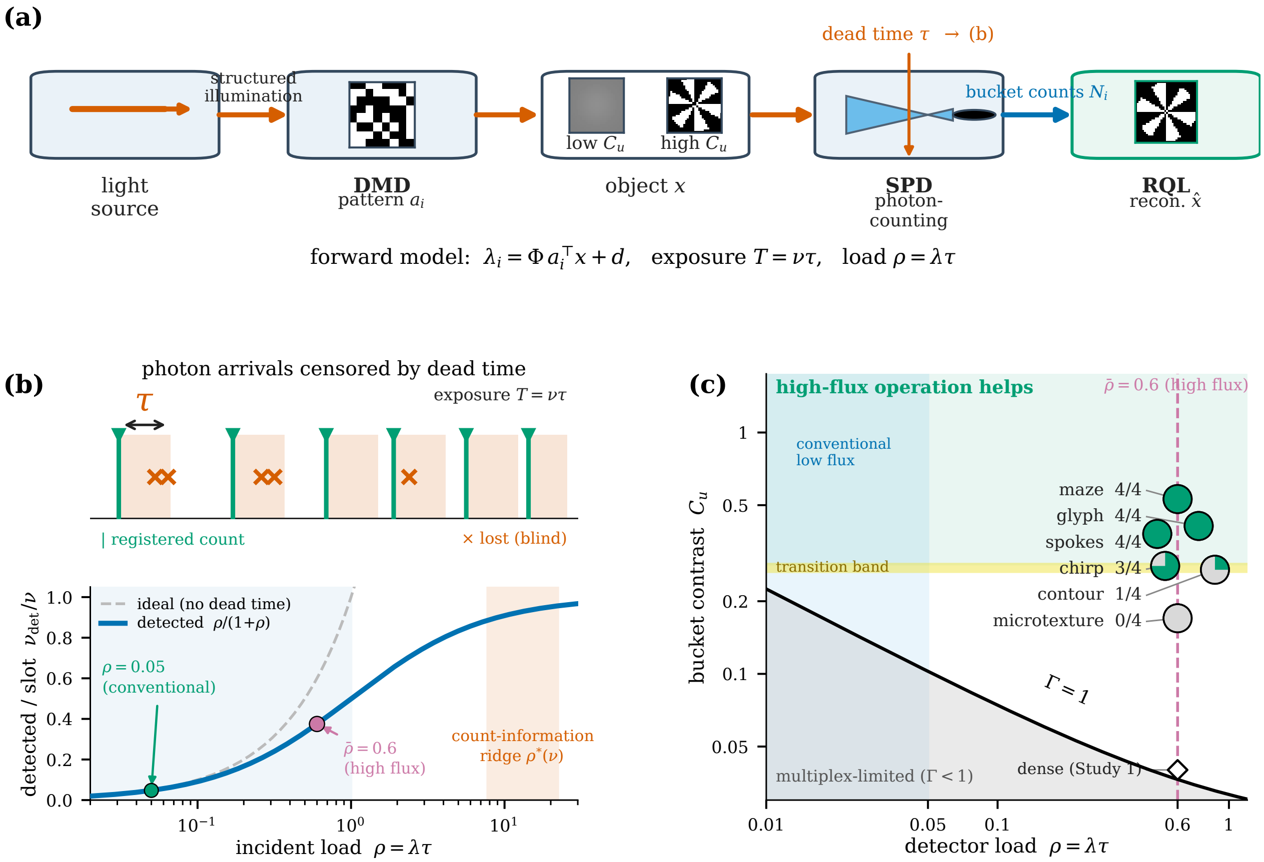

Figure 1: Contrast–dead-time operating map for high-flux single-pixel imaging. DMD: digital micromirror device; SPD: single-pixel (bucket) photon-counting detector; RQL: renewal quasi-likelihood reconstruction; a_i: binary illumination pattern; x: scene; N_i: integrated bucket count; $\hat{x}$: reconstruction; Φ: incident-flux scale; d: dark rate; τ: detector dead time; $\rho = \lambda\tau$: per-slot detector load; $\nu = T/\tau$: dead-time slots per exposure T; $C_u = \mathrm{sd}_i(u_i)/\mathrm{mean}_i(u_i)$: measured bucket-energy contrast of $u_i = a_i^\top x$; $\Gamma = C_u\sqrt{\nu\rho/(1+\rho)}$: contrast-to-count-noise index. (a) Single-pixel imaging chain: structured illumination from a DMD probes a low- or high-contrast scene x, and the bucket counts N_i feed the RQL reconstruction $\hat{x}$ under the forward model $\lambda_i = \Phi\, a_i^\top x + d$, exposure $T = \nu\tau$. (b) Dead-time mechanism: after each registered count the detector is blind for τ (shaded), censoring later arrivals ($\times$); the mean detected counts per slot $\nu_{\mathrm{det}}/\nu = \rho/(1+\rho)$ bend away from the ideal (dashed) as the load rises, while the exact count-information peaks on the ridge $\rho^*(\nu) = (6\nu)^{1/3} - 2/3$ ($\rho^* \approx 7.8$ and 22.2 at $\nu = 100$ and 2000, respectively), well beyond the conventional $\rho = 0.05$ and the high-flux $\bar{\rho} = 0.6$ operating points. (c) Operating map in the contrast–load plane: the photon-limited onset $\Gamma = 1$ separates the multiplex-limited region ($\Gamma < 1$, below) from the region where high-flux operation helps (above); the six Study-2 structural families are placed at the high-flux operating load $\bar{\rho} = 0.6$ by their measured bucket contrast and coloured by the frozen acquisition-speed gate outcome, with the number of accelerating instances shown as a green wedge: maze ($C_u = 0.53$), glyph (0.41) and spokes (0.38) accelerate 4/4, chirp (0.28) 3/4, contour (0.27) 1/4, and microtexture (0.17) 0/4, respectively, while the dense $k = 512$ rung ($C_u = 0.040$) sits at the $\Gamma = 1$ multiplex-limited boundary. All six family markers share the single operating load $\bar{\rho} = 0.6$; the horizontal offsets are cosmetic, applied only to separate near-coincident markers (chirp and contour differ by 0.01 in C_u), and thin leader lines tie each family label to its marker. The contrast-controlled transition band is descriptive and does not estimate a universal critical contrast. Grayscale object icons are for display purposes.

4 The finite-window count-information ridge

4.1 The missing-information identity

Let $I_N(\rho, \nu)$ denote the Fisher information of the single-frame count N with respect to $\log\lambda$, and $J = I_N/\nu$ its per-slot normalization ($J = 1$ is the saturation ceiling: one unit of log-rate information per dead-time slot). Writing $R_\nu \in [0, 1)$ for the residual dead-time phase left at the window end — the unobserved fraction of a dead-time period truncated by the frame boundary — the information splits as

$$I_N \;=\; \mathbb{E}[N] \;-\; \rho^2\, \mathbb{E}\big[\mathrm{Var}(R_\nu \mid N)\big]. \tag{3}$$

The first term is the complete-data information of the underlying renewal record (each registered event contributes one unit on the $\log \lambda$ scale); the second is the price of not observing the terminal phase: the frame boundary censors the last dead-time period, and the count alone cannot reveal how much of the window was spent in that final blind interval. The terminal phase has $P(R_\nu = 0) = 1/(1 + \rho)$ and density $\rho/(1 + \rho)$ on $(0, 1)$, so $\mathrm{Var}(R_\nu) \to \rho(\rho + 4)/[12(1 + \rho)^2] \to 1/12$ at high load — the censoring price grows as $\rho^2/(12\nu)$ per slot.

4.2 The cube-root law

Expanding J at large ρ and large ν,

$$J(\rho,\nu) = 1-\frac{1}{\rho}-\frac{\rho^2}{12\nu}+\frac{1}{\rho^2}-\frac{\rho}{6\nu}+\frac{\rho^4}{144\nu^2}+o(\nu^{-2/3}). \tag{4}$$

The leading tradeoff is between the saturation gap $1/\rho$ (a slot only carries full rate information when occupied by a completed detection–dead-time cycle) and the censoring price $\rho^2/(12\nu)$. The marginal balance $1/\rho^2 = \rho/(6\nu)$ gives $\rho^3 = 6\nu$, i.e. the information-optimal load

$$\rho^*(\nu) = (6\nu)^{1/3} - \tfrac{2}{3} + O(\nu^{-1/3}), \tag{5}$$

with peak per-slot information

$$J(\rho^*) = 1-\big(c^{-1}+c^2/12\big)\,\nu^{-1/3}+o(\nu^{-1/3}) = 1-0.8255\,\nu^{-1/3}+o(\nu^{-1/3}),\quad c = 6^{1/3}. \tag{6}$$

Exact evaluation of the renewal PMF gives $\rho^* = 4.53$, 6.16, 7.87, 9.99, 13.77, 17.45, and 22.16 at $\nu = 20$, 50, 100, 200, 500, 1000, and 2000, respectively; the closed form $(6\nu)^{1/3} - 2/3$ gives 4.27, 6.03, 7.77, 9.96, 13.76, 17.50, and 22.23 — a 5.9% gap at $\nu = 20$ shrinking to sub-percent by $\nu = 500$, consistent with the $O(\nu^{-1/3})$ remainder [Fig. 5(a)].

Two structural remarks. (i) The ridge is not a count-variance pinning effect: at $\rho = \rho^*$ the count variance $\mathrm{Var}\, N = \nu^{1/3}/6^{2/3}$ still diverges with ν; counts pin to the ceiling only at the much larger scale $\rho \asymp \sqrt{\nu}$. The information peak is an information-geometric tradeoff, not a saturation artifact. (ii) Beyond the ridge ($\rho \geq \rho^*$) the count information genuinely decreases — no estimator operating on counts, including the exact-likelihood reference, recovers it. High-flux operation is therefore beneficial only up to a computable boundary, and that boundary is one axis of the operating map reported here.

4.3 Deployment zoning and the CLT trust boundary

The Gaussian (CLT) surrogate information satisfies $J_{\text{exact}}/J_{\text{CLT}} \simeq 1 - \rho^2/(12\nu)$, so the surrogate stays within 10% of exact only for $\rho \leq \rho_{0.9} \asymp \sqrt{1.2\,\nu}$ — a $\sqrt{\nu}$ scaling, distinct from the $\nu^{1/3}$ ridge. We partition the (ρ, ν) plane into an RQL deployment zone (per-pattern $\rho_{95} \leq 1$, the regime the campaigns demonstrate), a transition band, an exact-reference mode (short-window and extreme-saturation reference computations with the exact PMF), and the information-decreasing region $\rho \geq \rho^*(\nu)$. Zoning uses the per-pattern 95th percentile of ρ, not its mean: spread-occupancy patterns distribute per-frame load, and the deployment guarantee must hold for the loaded tail, not the average.

4.4 Relation to prior work

Müller systematized renewal-theoretic analysis of extended and nonextended dead-time counters, relating distorted interarrival laws to the resulting count statistics and measurement-origin effects [17]. Yu and Fessler derived exact first and second moments for several dead-time counting models and used those moments to study intensity correction and the adequacy of Poisson reconstruction models [18]. Wang et al. derived pileup count statistics for idealized energy-discriminating nonparalyzable x-ray detectors and evaluated task-specific Cramér–Rao performance as a function of input rate, establishing substantial high-rate degradation of spectral-counting advantages [13]. Alvarez used renewal-process large-count moments and multivariate-Gaussian count models to quantify pileup-induced covariance, signal-to-noise loss, and material-estimation Cramér–Rao bounds [19]. Task-specific finite-rate optima also arise in nuclear counting; for example, Bécares and Blázquez optimized counting rate by balancing dead-time losses, counting uncertainty, source multiplicity, and dead-time-calibration uncertainty [14]. Grönberg et al. derived an exact finite-frame count probability mass function for a seminonparalyzable detector with nonzero pulse length, validated its moments against an electronic Monte Carlo model, and used the exact distribution together with Poisson and Gaussian approximations to evaluate Fisher information for a single-threshold soft-tissue-density task as a function of count rate; their fixed-frame Fisher curve exhibits an interior maximum followed by high-rate decline [20].

A complementary line retains richer event information. Rapp et al. modeled the sequence of dead-time-distorted detection times as a Markov chain, enabling accurate high-flux single-photon lidar and large acquisition-time reductions [8]. Jorgensen and Johnson derived asymptotic Fisher-information rates and efficient estimators for periodic dead-time event-detection processes, showing that activation and event statistics can retain information discarded by conventional histogram summaries [12].

Here we isolate a different detector-channel question: an ideal nonparalyzable detector is active at the start of a finite exposure, only its integrated scalar count is retained, and the parameter of interest is the log arrival rate itself. Treating $\rho = \lambda\tau$ and $\nu = T/\tau$ as independent axes, we derive the exact finite-window information surface, the terminal-phase missing-information identity of Eq. (3), and the principal information ridge of Eq. (5). These statements concern the active-start ideal count-only channel and do not replace task-specific finite-rate optima for nonzero-pulse-length detectors or information bounds for time-resolved dead-time event records.

5 Study 1: dense patterns — the multiplex-limited phase

5.1 Preregistered design

We tested whether renewal quasi-likelihood reconstruction permits operation at $\bar{\rho} = 0.6$ with a statistically significant reduction in total optical integration time relative to the conventional $\bar{\rho} = 0.05$ operating point, on 64^2 scenes under dense Bernoulli illumination (occupancy 50%). Scenes are the units of inference: 24 confirmatory detail targets (six structural families × four instances), each measured at nine dwell points $\nu \in \{5, \ldots, 2000\}$ with five measurement seeds. Per scene, seed-mean radiometric PSNR versus optical integration time is isotonically regressed for each arm; the quality target is Q_{90}, the safe-arm minimum plus 90% of the safe-arm span, and the acquisition-speed ratio S_j is the ratio of safe to high-flux crossing times, with a frozen censoring taxonomy (a safe-arm span below 0.5 dB is uninformative; a high-flux arm that never reaches Q_{90} scores $S_j = 0$). The confirmatory gate — median $S \geq 3$, family-stratified nested-bootstrap (10 000 replicates) lower bound > 1, and $\geq 18/24$ scenes with $S_j > 1$ — and every constant above were frozen in an immutable commit before any confirmatory reconstruction existed, together with a fixed-dwell secondary endpoint (median ≥ 1 dB at the terminal dwell, lower bound > 0, $\geq 18/24$ positive).

5.2 Preregistered outcome

The confirmatory time-to-quality gate returned negative: in 24/24 scenes the safe-arm quality span across the entire dwell grid fell below the preregistered 0.5 dB informativeness floor (SAFE_RANGE_UNINFORMATIVE; $S_{\text{gate}} = 1$ by the frozen censoring taxonomy), so no time-to-quality acceleration could be certified. The preregistered fixed-dwell secondary, in contrast, was uniformly positive but small: at $\nu = 2000$ the high-flux arm gained a median +0.28 dB (radiometric PSNR), positive in 24/24 scenes, with family-stratified bootstrap lower bound +0.23 dB — statistically significant, but far below the preregistered 1 dB practical-relevance bar. Both outcomes are reported as frozen; no endpoint, gate, or analysis constant was altered after the freeze.

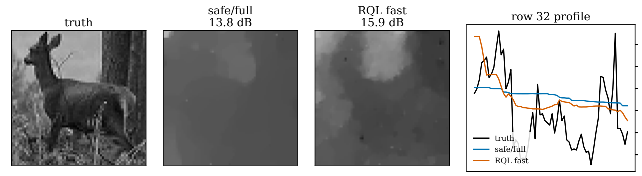

Figure 2: Natural-cohort representative: truth, safe operating point at full dwell (13.8 dB), and high-flux RQL at equal dwell (15.9 dB). Display subject selected by rule (disclosed): the photograph whose fixed-dwell gain lies closest to the cohort median of +1.44 dB (`stl_16`, +1.46 dB) — a median representative, not a best case. The cohort is prior-limited at the studied photon budget: reconstructions are spatially smoothed and the high-flux benefit is radiometric, visible both in the recovered coarse structure and in the center-row intensity profile (right): the safe arm is nearly flat while the high-flux arm tracks the scene's intensity excursions at equal dwell.

5.3 Natural-scene cohort (secondary, no gate)

On the preregistered natural-image cohort (30 scenes, same dense-pattern protocol, no confirmatory gate), the acquisition-speed endpoint was safe-range-uninformative for 12/30 scenes, while the fixed-dwell gain was larger and more uniform than on the detail cohort: median +1.44 dB, 29/30 scenes positive, family-stratified bootstrap lower bound +1.18 dB. For context, classical correlation ghost imaging on the identical measurement records scores 4.8–6.3 dB — at this photon budget (≈ 95 detected photons per pixel over the full acquisition) the likelihood-based reconstruction, not the sampling ratio, is what renders the data usable. Smooth, prior-limited scenes thus occupy the low-benefit region of the acquisition-speed map even where the fixed-dwell radiometric benefit of high-flux operation is substantial.

5.4 Mechanism: bucket multiplex contrast

The two outcomes are jointly explained by the bucket contrast of dense multiplexed illumination. For occupancy-1/2 patterns on n pixels the bucket energies $u_i = a_i^\top x$ concentrate as $C_u = \mathrm{sd}(u)/\mathrm{mean}(u) \sim 1/\sqrt{n}$ ($\approx 1.6\%$ at 64^2): the structural information rides on a relative modulation of order 10^{-2}, so the per-count information about *structure* is contrast-limited, not count-limited — raising the flux adds photons to a nearly constant bucket. This is the multiplex-limited regime of the operating map; it motivates Study 2.

6 Study 2: the contrast ladder — equal load, equal dose

6.1 Preregistered design

Study 2 varies spatial occupancy $k \in \{512, 32, 16, 1\}$ (occupancy 50%, 3.1%, 1.6%, 0.1% of $n = 1024$ pixels at 32^2) using binary matrices with exact row and column sums k, while holding the *mean detector load and the incident photon dose fixed across the ladder*: the reconstruction operator is $A^{(k)} = (n/k)\,B^{(k)}$ at flux $\Phi = \bar{\rho}/\tau$, so every occupancy shares identical mean count rate, identical incident photons per pattern, and identical cumulative dose per object pixel; only the on-pixel peak intensity rises as n/k. Sparse patterns therefore cannot win by starving the detector — the mechanism prior sparse-pattern work uses to mitigate pile-up [15, 16]. Only the $k = 16$ arm carries the confirmatory gate (constants unchanged from Study 1); $k = 32$ is a preregistered robustness arm and $k = 512$ and $k = 1$ are controls, all without gates. Two mechanism variables are recorded descriptively and never enter any gate: the measured bucket-energy contrast $C_u = \mathrm{sd}_i(u_i)/\mathrm{mean}_i(u_i)$ and the contrast-to-count-noise index $\Gamma = C_u\sqrt{\nu\rho/(1+\rho)}$, whose crossing of unity marks the onset of the photon-limited structural regime. A contrast-to-noise ratio on generator-frozen signal/background regions is carried as a no-gate co-secondary, and all photon budgets are reported on two axes: the measured-photons axis $S_{\mathrm{det}} = (Mn)^{-1}\sum_i N_i$ (the convention of Ref. [4], whose experiments span $S_{\mathrm{det}} = 0.096$–0.44) and the incident axis S_{inc}, which differ nonlinearly under dead time.

6.2 The measured contrast ladder

The measured bucket contrast follows the designed ladder: $C_u = 0.040$, 0.222, 0.316, and 1.270 at $k = 512$, 32, 16, and 1, respectively, and the contrast-to-count-noise index at the high-flux operating point ($\bar{\rho} = 0.6$, $\nu = 2000$) crosses the photon-limited threshold on every sparse rung: $\Gamma = 1.09$, 6.07, 8.66, and 34.79, respectively [Fig. 5(c)].

6.3 Primary confirmatory outcome ($k = 16$)

The preregistered acquisition-speed gate returned formally negative: of its three frozen criteria, the effect-size criteria passed — median $S_{\mathrm{gate}} = 6.8$ (bar: ≥ 3) with family-stratified bootstrap lower bound 2.8 (bar: > 1) — but the breadth criterion failed, with 16/24 scenes accelerating against the preregistered 18/24 bar [Fig. 5(e)]. The frozen fixed-dwell secondary passed all three of its criteria (median +1.16 dB at the terminal dwell, 19/24 scenes positive, bootstrap lower bound +0.72 dB); by the preregistered hierarchy it cannot rescue the failed primary, and no geometry, image-class, or endpoint redesign follows.

The composition of the failure is structured. The eight non-accelerating scenes comprise the entire microtexture family (0/4; three safe-range-uninformative, one fast-arm-censored), three of four contour instances, and one chirp instance, while glyph, maze, and spokes accelerated 4/4 with per-scene speedups of 5–33×. After unblinding, a descriptive family-level analysis showed that the six family success fractions were monotonically ordered by their corresponding mean measured bucket contrasts ($C_u = 0.53$ maze 4/4, 0.41 glyph 4/4, 0.38 spokes 4/4, 0.28 chirp 3/4, 0.27 contour 1/4, 0.17 microtexture 0/4). This six-point ordering was not preregistered, did not enter any decision rule, and does not estimate a universal critical contrast; it is consistent with the confirmatory cohort spanning a contrast-controlled transition.

6.4 Fixed-dwell gain along the ladder

At fixed dwell ($\nu = 2000$, equal optical integration time and equal incident dose across occupancies) every rung of the ladder shows a positive high-flux median gain, and the dense rung is the smallest: per-image median +0.64 dB (24/24 scenes positive) at $k = 512$, +4.13 dB (24/24) at $k = 32$, +1.16 dB (19/24) at the gate-carrying $k = 16$ arm, and +3.56 dB (21/24) at the raster control, respectively [Fig. 5(e)]. The profile across the sparse rungs is not monotone in contrast: sparsity raises bucket contrast but simultaneously raises the operator conditioning burden ($\kappa_A = n/k$), and the two effects compete — $k = 32$ combines most of the contrast benefit with half the conditioning burden of $k = 16$ and attains the largest gain. The dense-rung value reproduces the Study-1 magnitude, confirming that the 64^2-to-32^2 re-parameterization did not manufacture the effect.

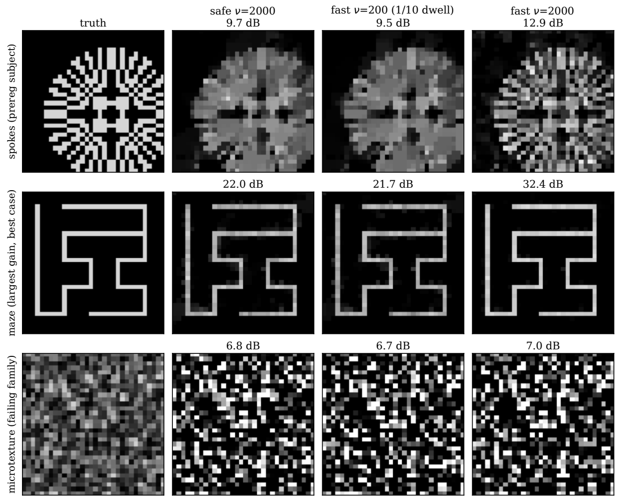

Figure 3: Study-2 reconstructions at the gate-carrying occupancy ($k = 16$, RQL, seed 0), all through the frozen production path. Row subjects by disclosed rule: the preregistered display subject (spokes), the largest fixed-dwell gain in the frozen verdict table (maze, +9.7 dB seed-mean — a best case, labeled as such), and a failing-family representative (microtexture). Columns: truth; safe operating point at full dwell; high flux at one tenth of the dwell; high flux at full dwell. On contrast-adequate scenes the high-flux arm matches the safe arm's quality in a tenth of the acquisition time and far exceeds it at equal time (22.0 → 32.4 dB on the maze), while the low-contrast microtexture is indifferent to flux at every dwell — the operating map in three rows.

6.5 Mechanism separation: equal-load gain is not rate starvation

On the six frozen mechanism representatives we ran the $k = 16$ geometry in two flux conventions: (i) *normalized* — the Study-2 mechanism, equal mean load and equal dose; and (ii) *unnormalized* — sparsity is allowed to reduce the detector count rate, as in prior sparse-pattern mitigation. In the normalized convention the high-flux arm retains a positive gain of +1.81 and +1.10 dB at $\nu = 200$ and 2000, respectively, at detector loads of $S_{\text{det}} = 0.072$–0.72 counts per pattern per pixel. The unnormalized convention shows larger *relative* deltas but strictly lower absolute quality at every dwell (e.g. 7.45 versus 10.26 dB at the safe point, $\nu = 2000$): it wins its comparison by starving the detector. The equal-load, equal-dose gain is therefore a contrast effect, not a rate-reduction effect.

6.6 Dynamic scattering

Under multiplicative dynamic scattering (lognormal AR(1) factors, $\mathrm{CV}(\alpha) = 0.20$, applied before dead time) the high-flux gain persists and grows with dwell: +0.66, +1.84, and +2.32 dB at $\nu = 20$, 200, and 2000, respectively [Fig. 5(f)], reported on the measured-photons axis S_{det} for direct comparison with single-photon underwater single-pixel experiments [4].

6.7 Robustness to detector mismatch

Detector-mismatch consequences were quantified descriptively on the 64^2 cohort at the preregistered anchor ($M/n = 0.5$, $\nu = 500$), pairing each perturbed cell with its no-mismatch baseline. Every anchor perturbation leaves the RQL radiometric PSNR within 0.34 dB of baseline at both operating points: dead-time miscalibration $\hat{\tau} = \tau(1 \pm 0.1)$ costs at most 0.12 dB (with a ±6% radiometric flux bias, the largest observed); a 10% dark rate costs 0.16 dB when calibrated and 0.34 dB when unmodeled; afterpulsing at 2–5% costs at most 0.09 dB. Under the same perturbations the pre-correction arm degrades more in five of six anchors (up to 0.79 dB, unmodeled dark). Start-mode, guard-time, and dead-time-jitter axes, severity sweeps to 50% dark, and the descriptive measurement-audit distributions are tabulated in the supplement.

6.8 Exact-likelihood reference at small scale

Because RQL is a convex surrogate, we verified at 8^2 and 16^2 that it tracks the exact renewal likelihood under an identical TV functional and the identical frozen selection rule: across the full preregistered grid ($\rho \in \{0.03, \ldots, 2\}$, $\nu \in \{20, \ldots, 2000\}$, three seeds, six scenes, 864 paired reconstructions) the median exact-minus-RQL radiometric-PSNR gap is ≤ 0.01 dB throughout the deployment zone and reaches only -0.05 dB in the extreme-load corner (16^2, $\rho \geq 0.6$, $\nu = 2000$). Individual cells differ in both directions (up to 9 dB), consistent with the nonconvexity of the exact objective, with no systematic advantage to either arm; full tables and multistart diagnostics are provided in the supplement. The convex surrogate therefore costs essentially nothing in reconstruction quality while guaranteeing a unique optimum.

7 Discussion

The scientific position established by the two studies is deliberately mixed. Dense illumination is multiplex-limited; sparse equal-load illumination produces large and statistically supported benefits for many, but not sufficiently many, preregistered scenes to pass the breadth gate. The scene response is nevertheless structured by bucket contrast, while the tested occupancy ladder exposes an independent conditioning penalty.

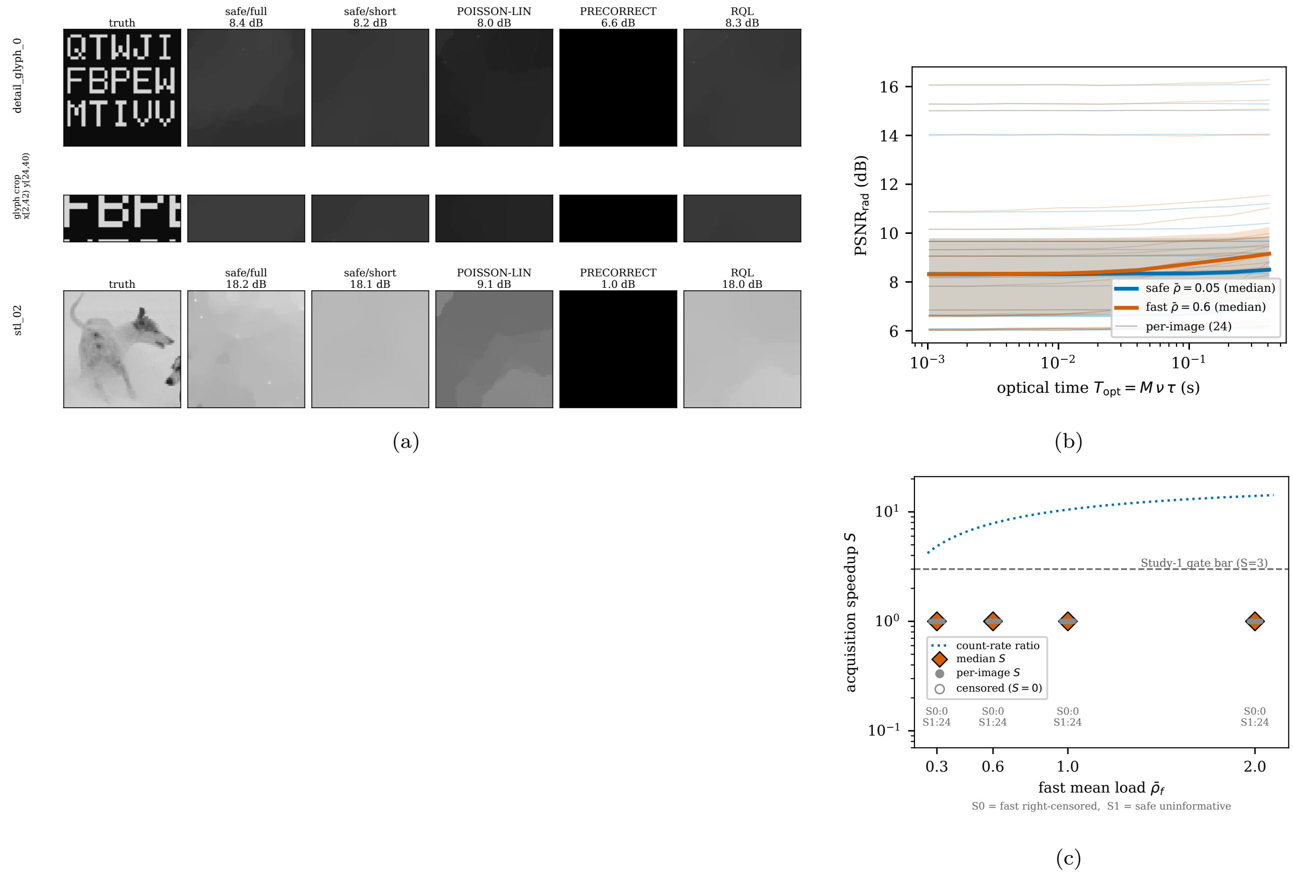

Figure 4: Study 1: the dense-pattern multiplex-limited regime (frozen preregistered display subjects). (a) Image arrays for the confirmatory detail subject `detail_glyph_0` (top, with the frozen zoom crop) and the natural-cohort boundary subject `stl_02` (bottom): truth, safe operating point at full and short dwell, and the three high-flux arms at $\bar{\rho} = 0.6$, $\nu = 50$. Dense multiplexing leaves the detail subject prior-limited in every arm (the frozen negative made visible), while on the natural subject the naive and pre-corrected arms collapse (9.1 and 1.0 dB) and RQL retains full quality (18.0 dB). (b) Seed-mean radiometric PSNR versus optical integration time for all 24 detail scenes (thin lines) with cohort medians (thick): the safe-arm median spans less than 0.2 dB — the safe-range-uninformative verdict. (c) Per-scene acquisition-speed bounds versus fast-arm load: every scene sits at the $S = 1$ censoring bound, below the preregistered gate bar, while the count-rate ratio (dotted) shows what the detector alone would permit — the gap between the two is the multiplex limitation.

Together with the exact count-information ridge and the passed fixed-dwell secondary, this constitutes an operating map: a computable answer to when high-flux operation is worth engaging, rather than a universal acceleration claim.

Two scope boundaries must be stated plainly. First, all results are simulation-based; the exact renewal forward model, the detector-mismatch envelopes of Section 6.7, and the openly released frozen benchmark compensate but do not substitute for optical-bench validation. Second, sparse operation at equal dose concentrates the same illumination into fewer pixels: the on-pixel peak irradiance rises as n/k ($64\times$ at the gate-carrying occupancy), which spatial-light-modulator power budgets, laser-safety limits, and local photodamage constraints may cap before the detector does. The operating map therefore still requires optical-bench validation, and its sparse rungs are conditional on peak-power headroom.

Three extensions follow directly from the failure structure. For uniformly scattered fixed-k supports, $C_u^2 = \frac{n-k}{k(n-1)}\,(n\|x\|_2^2 - 1)$, giving the observed $k^{-1/2}$ spatial-averaging law. A local support instead averages only the effective number of independent correlation cells within the patch and may therefore preserve modulation from scale-matched structure. Local, foveated and multiscale single-pixel patterns have previously been developed for resolution and measurement efficiency; optimizing their spatial scale for equal-load dead-time contrast is a distinct future design problem.

The present results do not establish that adaptive illumination is necessary. Although dead-time nonlinearity makes the statistically efficient pattern scale scene-dependent, a strict adaptive advantage requires a specified scene class, resource constraint and loss. A natural follow-up is to seek a minimax or Chernoff-information separation between fixed multiscale illumination and a two-stage scale-estimation-plus-matched-pattern policy.

A coherent extension would use a coarse complex-field estimate to synthesize an optical displacement that suppresses the predicted common mode before photon counting. Unlike electronic balanced subtraction, a dark-port receiver removes photons before dead-time loss. Exact nulling, however, gives quadratic residual sensitivity and demands spatial-mode matching and phase stability; the relevant operating variable is therefore an optimized displacement bias rather than maximal null depth. Testing this receiver first under static and then dynamic scattering is a natural optical-bench extension. More broadly, architectures that retain per-photon timing [7, 8] occupy a different information regime than the count-only channel analyzed here; the map presented in this work is the count-only baseline against which such richer architectures can be costed. The proposed operating map, the deployment-legal reconstruction rule, and the frozen open benchmark open up an avenue for dwell-time-limited and photon-budget-limited single-pixel systems operating deep in the dead-time regime.

Code and data availability

All simulation code, frozen preregistration documents, per-cell manifests, and evidence bundles are available at [repo URL wording — user decision].

Acknowledgments

[funding/acknowledgments — user decision]

References

[1] Marco F. Duarte, Mark A. Davenport, Dharmpal Takhar, Jason N. Laska, Ting Sun, Kevin F. Kelly, and Richard G. Baraniuk. Single-pixel imaging via compressive sampling. *IEEE Signal Processing Magazine*, 25(2):83–91, 2008.

[2] Jeffrey H. Shapiro. Computational ghost imaging. *Physical Review A*, 78(6):061802, 2008.

[3] Matthew P. Edgar, Graham M. Gibson, and Miles J. Padgett. Principles and prospects for single-pixel imaging. *Nature Photonics*, 13(1):13–20, 2019.

[4] Yining Hao, Yang Peng, Tianshun Zhang, and Wen Chen. High-quality photon-limited imaging through dynamic and complex scattering media with a single-photon detector. *APL Photonics*, 10(10):106119, 2025.

[5] Yining Hao, Yin Xiao, and Wen Chen. Single-pixel imaging through random media with automated adaptive corrections. *Applied Physics Letters*, 126(13):131105, 2025.

[6] Yining Hao and Wen Chen. Single-pixel complex-field imaging through scattering media. *Optics Letters*, 50(6):1949–1952, 2025.

[7] Joshua Rapp, Yanting Ma, Robin M. A. Dawson, and Vivek K. Goyal. Dead time compensation for high-flux ranging. *IEEE Transactions on Signal Processing*, 67(13):3471–3486, 2019.

[8] Joshua Rapp, Yanting Ma, Robin M. A. Dawson, and Vivek K. Goyal. High-flux single-photon lidar. *Optica*, 8(1):30, 2021.

[9] Anant Gupta, Atul Ingle, Andreas Velten, and Mohit Gupta. Photon-flooded single-photon 3d cameras. In *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 6763–6772, 2019.

[10] Anant Gupta, Atul Ingle, and Mohit Gupta. Asynchronous single-photon 3d imaging. In *2019 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 7908–7917, 2019.

[11] Xinyuan Liu, Yue Ma, Song Li, Jian Yang, Zhiyu Zhang, and Xin Tian. Photon counting correction method to improve the quality of reconstructed images in single photon compressive imaging systems. *Optics Express*, 29(23):37945, 2021.

[12] Frederic J. N. Jorgensen and Steven G. Johnson. Fundamental bounds and efficient estimation for dead-time-constrained event detection, with application to single-photon lidar, 2026. arXiv:2605.23210.

[13] Adam S. Wang, Daniel Harrison, Vladimir Lobastov, and J. Eric Tkaczyk. Pulse pileup statistics for energy discriminating photon counting x-ray detectors. *Medical Physics*, 38(7):4265–4275, 2011.

[14] V. Bécares and J. Blázquez. Detector dead time determination and optimal counting rate for a detector near a spallation source or a subcritical multiplying system. *Science and Technology of Nuclear Installations*, 2012:240693, 2012.

[15] Shuai Liu, Xu-Ri Yao, Xue-Feng Liu, Da-Zhi Xu, Xing-Da Wang, Bing Liu, Chao Wang, Guang-Jie Zhai, and Qing Zhao. Pile-up effect in an infrared single-pixel compressive lidar system. *Optics Express*, 27(16):22138–22146, 2019.

[16] Yuyuan Han, Peng Wu, Hanxiong Xu, Bin Li, Yuchen He, Jianbin Liu, Hui Chen, and Huaibin Zheng. Generalized sparse hadamard single-pixel imaging. *Optics Express*, 33(20):43324–43341, 2025.

[17] Jörg W. Müller. Dead-time problems. *Nuclear Instruments and Methods*, 112(1–2):47–57, 1973.

[18] Daniel F. Yu and Jeffrey A. Fessler. Mean and variance of single photon counting with deadtime. *Physics in Medicine and Biology*, 45(7):2043–2056, 2000.

[19] Robert E. Alvarez. Signal to noise ratio of energy selective x-ray photon counting systems with pileup. *Medical Physics*, 41(11):111909, 2014.

[20] Fredrik Grönberg, Mats Danielsson, and Martin Sjölin. Count statistics of nonparalyzable photon-counting detectors with nonzero pulse length. *Medical Physics*, 45(8):3800–3811, 2018.

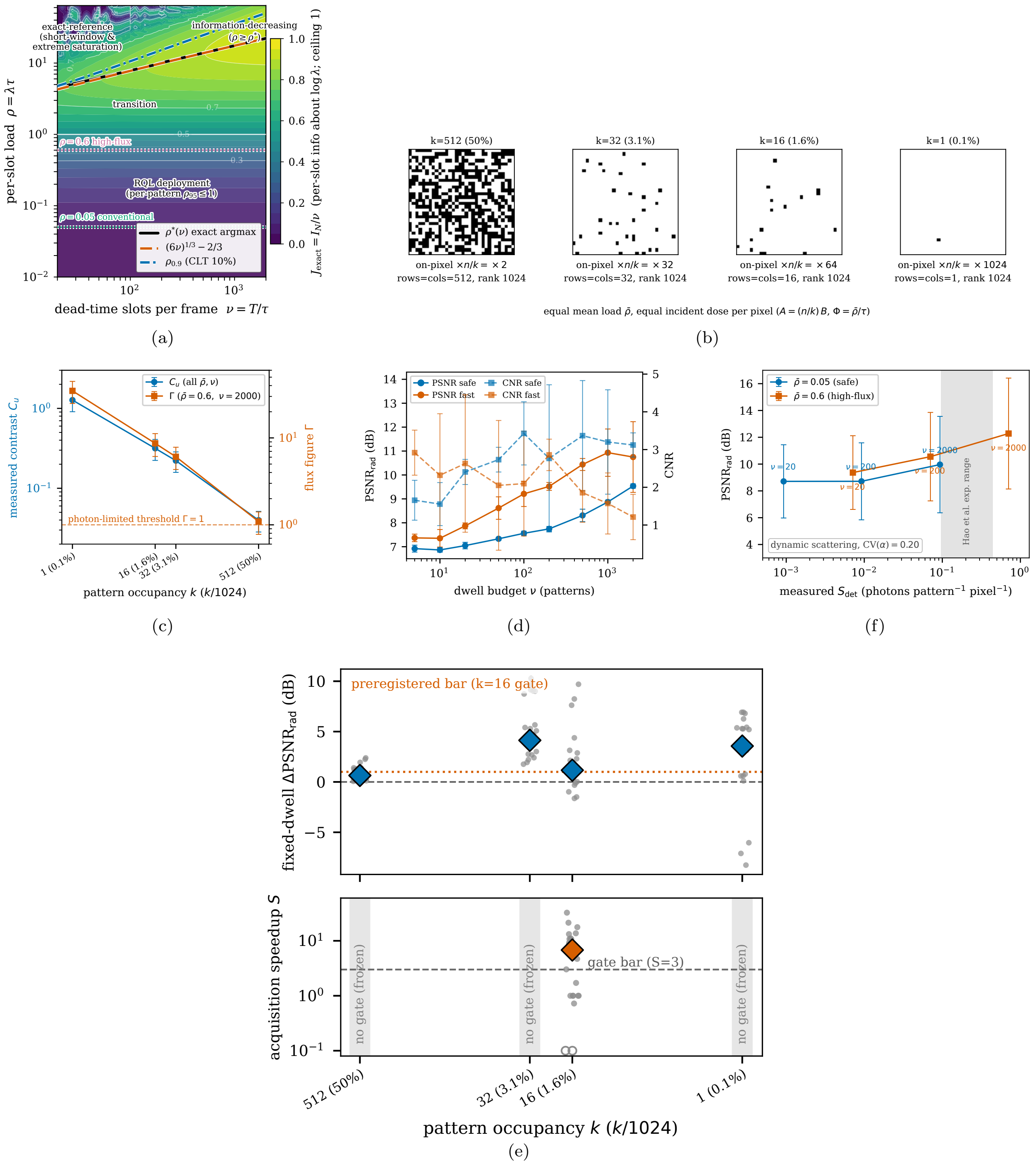

Figure 5: The contrast–dead-time operating map (Study 2). (a) Exact count-Fisher-information map with the cube-root ridge $\rho^*(\nu)$. (b) Representative occupancy-ladder masks ($k = 512/32/16/1$) under equal mean detector load and equal incident dose ($A = (n/k)B$, $\Phi = \bar{\rho}/\tau$). (c) Measured bucket contrast C_u and photon-limited index Γ versus occupancy; the dense rung sits at the $\Gamma = 1$ threshold. (d) Radiometric PSNR and CNR versus dwell for the preregistered resolution-target subject (`detail32_spokes_0`, $k = 16$); the ridge of panel (a) is a scalar count-information result and no coincidence with the image optimum is claimed. (e) Fixed-dwell gain by occupancy (top; the dotted line is the $k = 16$ preregistered bar) and per-scene acquisition speedups on the gate-carrying $k = 16$ arm (bottom); open markers are fast-arm-censored scenes. (f) Dynamic-scattering secondary in measured-photons units; the shaded band is the experimental range of Ref. [4].